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Q => We define $f(x,y)$ as number of different corresponding bits in Binary representation of X and Y. For example $f(2,7)=2$, since the binary representation of 2 and 7 are 010 and 111 respectively. The first and third bit differ so $f(2,7)=2$.

You are given an array of N +ve integers A_1, A_2 —— A_N. Find sum of $f(A_i, A_j)$ for all pairs (i,j) such that $1 \le i, j \le N$. Return the answer modulus 10^9+7.

Sol => Bruteforce

Algorithm
1-) Check every pair (i,j)
2-) count different bits using XOR
If $x \wedge y$ has k set bits, the k bits differ

Complexity => $O(N^2)$

Optimized

Code=>
```
# include <iostream>
using namespace std;
int func (vector <int> & arr) {
    const int MOD = 1e9+7;
    long long ans = 0;
    int n = A.size();
```

```
for (int bit=0; bit<32; bit++){
    long long ones=0;
    for(int i=0; i<n; i++){
        if (A[i] & (1<< bit))  ones++;
    }

    long long zeroes = n- ones;
    ans = (ans + (ones* zeroes% MOD)* 2 %MOD)
                                        %MOD
}
return ans;
}
```

<u>Time complexity</u> : $O(N)$

<u>Space complexity</u> : $O(1)$

<u>Algorithm</u> :

1.) Initialize ans = 0

2.) for bit = 0 to 31 :

a.) Count ones in array having that bit set

b.) zeroes = N - ones

c.) Add ones * zeroes * 2 to answer (mod 10^9+7)

3.) Return ans.